In this section I show three ways to create custom middleware components, as well as how to create branches in your middleware pipeline where a request can flow down either one branch or another. By combining the methods demonstrated in this section, you'll be able to create custom solutions to handle your specific requirements.

We start by creating a middleware component that returns the current time as plain text whenever the app receives a request. From there we'll look at branching the pipeline, creating general-purpose middleware components, and encapsulating your middleware into standalone classes. Finally, in section 31.1.5 you'll see how to turn your custom middleware component into an endpoint and integrate it with the endpoint routing system.

31.1.1 Creating simple apps with the Run extension

As you've seen in previous chapters, you define the middleware pipeline for your app in Program.cs by adding middleware to a `WebApplication` object, typically using extension methods, as in this example:

```
WebApplicationBuilder builder = WebApplication.CreateBuilder(args);
WebApplication app = builder.Build();
app.UseExceptionHandler();
app.UseStaticFiles();
app.Run();
```

When your app receives a request, the request passes through each middleware component, each of which gets a chance to modify the request or handle it by generating a response. If a middleware component generates a response, it effectively short-circuits the pipeline; no subsequent middleware in the pipeline sees the request. The response passes back through the earlier middleware components on its way back to the browser.